

Satellite Gateway

Satellite gateway is a ground station that interfaces between the satellite and the local area network (LAN). It houses the antennas and equipment that convert the radio frequency (RF) signals to internet protocol (IP) signals and vice versa for internet access and data transfer. Voice information can also be carried using Voice over IP protocol. It provides connectivity to the terrestrial backhaul network with satellite, converts baseband data into RF signals for transmission to satellite, and sends the received RF signals back to baseband for decoding the data information in various forms (voice, image, video, text) as desired.



(Source: <https://x2n.com>)

optic cables), enabling the users to communicate directly with the satellites.

OneWeb

OneWeb, backed by the British government, Bharti Enterprises, Eutelsat, SoftBank, Hughes Networks, and Hanwha, is planning to launch a total of 648 satellites to provide internet services with a download speed of 165Mbps, upload speed of 30Mbps, and latency of 45 milliseconds in regions located above 50 degrees north latitude, which include Alaska, Canada, Greenland, UK, and Northern Europe.

The Indian Space Research Organisation (ISRO) is set to launch 36 OneWeb satellites from the Satish Dhawan Space Centre, Sriharikota, adding to the existing constellation of 582 satellites to enable OneWeb to launch services in India. By the end of the year, OneWeb plans to launch the remaining satellites to provide services across the globe.

Viasat

Viasat operates ViaSat-1, Wild-Blue1, Anik-F2, and ViaSat-2, which are much larger than the Starlink satellites and operate at a higher orbit, providing services to about 590,000 US subscribers. Presently,

it offers a download speed ranging from 12Mbps to 100 Mbps, which is expected to increase with the next satellite Viasat-3 (scheduled to be launched on April 18, 2023).

Telesat

Currently, Telesat—a Canadian company, is providing internet services to business houses using 15 geostationary satellites. It is planning to launch a new constellation, Lightspeed, consisting of 1,600 LEO satellites to provide satellite internet services.

The first group of satellites is scheduled to be launched in 2023 with the availability of the connections expected in 2024. Telesat is

planning to provide speeds of the order of GBs and latency at par with fibre networks.

Geespace

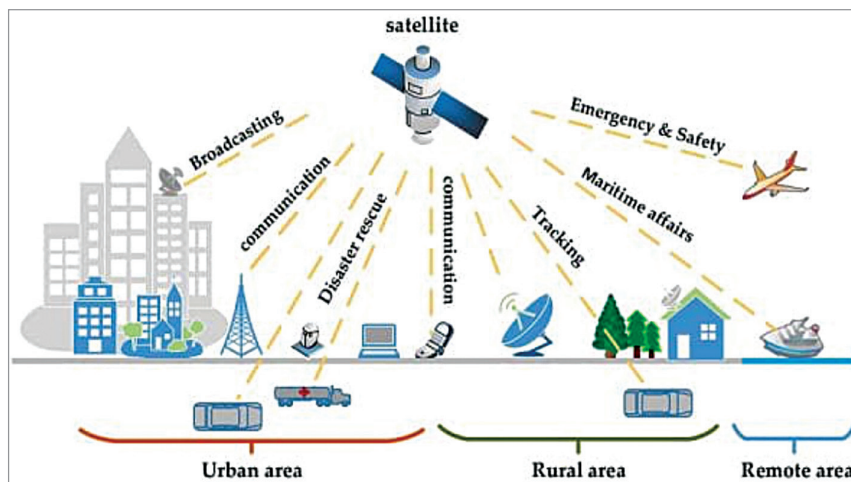
The Geely Technology Group formed Geespace in 2018 for developing, launching, and operating a constellation, Geely Future Mobility, consisting of 240 LEO satellites to provide high-speed internet services and support highly precise navigation and cloud computing capabilities across China. In June 2022, Geespace launched the first batch of nine satellites.

The first phase to place 72 satellites in orbit is planned to be completed by 2025. Geely Technology Group is the first privately owned enterprise in China to develop, produce, and operate low-orbit commercial satellites.

Amazon

Amazon is planning to invest more than \$10 billion to scale up Project Kuiper and Amazon Web Services (AWS) to leverage their global logistics and operations, as well as provide satellite services commercially. Under these projects, the United Launch Alliance, Arianespace, and Blue Origin together have launched 3,236 satellites into LEO.

In 2020, Amazon unveiled a



Remote area access is one of the major benefits of satellite Internet (Source: <https://www.rfpage.com>)